

AQI Forecasting Model Evaluation

Effect of Train/Validation Split Ratio on Random Forest, Extra Trees, and Ensemble Model Performance

1. Dataset Overview

The dataset used for this evaluation contains 23,257 total samples, each represented as a 72-timestep by 16-feature sequence (shape: $23257 \times 72 \times 16$), flattened to a 1,152-dimensional input vector for the machine learning models. Each sample is paired with 3 target values, corresponding to Air Quality Index (AQI) forecasts for Day 1, Day 2, and Day 3 ahead.

Three separate experiments were run, each using a different train/validation split ratio of the same underlying 23,257-sample dataset, to evaluate how split ratio affects model performance across Random Forest, Extra Trees, and an Ensemble of the two.

2. Experiment 1 — 80% Train / 20% Validation Split

Data Split Details

Subset	Samples	Percentage	Total Dataset
Training set	18,605	80.00%	23,257
Validation set	4,652	20.00%	

Validation Results

Model	Forecast Day	R ²	MAE	RMSE
Random Forest	Day 1	0.7838	8.6362	11.0180
Random Forest	Day 2	0.4988	12.9791	16.7747
Random Forest	Day 3	0.3719	14.3517	18.7527
Extra Trees	Day 1	0.7389	9.4948	12.1060
Extra Trees	Day 2	0.4641	13.4985	17.3455
Extra Trees	Day 3	0.3503	14.6915	19.0732
Ensemble	Day 1	0.7649	9.0151	11.4888
Ensemble	Day 2	0.4852	13.1954	17.0008
Ensemble	Day 3	0.3644	14.4867	18.8646

3. Experiment 2 — 70% Train / 30% Validation Split

Data Split Details

Subset	Samples	Percentage	Total Dataset
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Training set	16,279	70.00%	23,257
Validation set	6,978	30.00%	

Validation Results

Model	Forecast Day	R ²	MAE	RMSE
Random Forest	Day 1	0.8664	7.9575	10.3920
Random Forest	Day 2	0.6576	12.6354	16.6409
Random Forest	Day 3	0.5415	14.6618	19.2588
Extra Trees	Day 1	0.8364	8.7417	11.5026
Extra Trees	Day 2	0.6397	13.0540	17.0707
Extra Trees	Day 3	0.5261	15.1242	19.5796
Ensemble	Day 1	0.8544	8.2723	10.8516
Ensemble	Day 2	0.6523	12.7773	16.7683
Ensemble	Day 3	0.5387	14.8410	19.3175

4. Experiment 3 — 60% Train / 40% Validation Split

Data Split Details

Subset	Samples	Percentage	Total Dataset
Training set	13,954	60.00%	23,257
Validation set	9,303	40.00%	

Validation Results

Model	Forecast Day	R ²	MAE	RMSE
Random Forest	Day 1	0.8429	8.0929	10.5517
Random Forest	Day 2	0.6018	12.8438	16.7965
Random Forest	Day 3	0.4761	14.7588	19.2544
Extra Trees	Day 1	0.8003	9.0695	11.8956
Extra Trees	Day 2	0.5647	13.3950	17.5617
Extra Trees	Day 3	0.4399	15.3342	19.9076
Ensemble	Day 1	0.8252	8.5117	11.1285
Ensemble	Day 2	0.5876	13.0503	17.0938

Ensemble	Day 3	0.4633	14.9984	19.4870
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5. Experiment 4 — LSTM + Random Forest Custom Model (70% Train / 30% Validation Split)

This experiment uses a custom hybrid architecture combining an LSTM (for temporal feature extraction) with a Random Forest (for final prediction), rather than the standalone Random Forest / Extra Trees / Ensemble models used in Experiments 1–3. Train/validation split sample counts were not provided for this run, so the split details table is omitted here — only final validation performance is reported below.

Validation Results

Model	Forecast Day	R ²	MAE	RMSE
LSTM + Random Forest (custom)	Day 1	0.8765	7.6541	9.9950
LSTM + Random Forest (custom)	Day 2	0.6513	12.8326	16.7927
LSTM + Random Forest (custom)	Day 3	0.5239	15.2170	19.6235
